

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 01: Hotel Booking System - Attribute & Entity Dictionary

[PASTE: Exp 1: Left Page (Entity Relational Design)]

Entity Name | Primary Key | Foreign Keys | Attributes & Data Domains

Guests: guest_id (INT) -> first_name, last_name, email, phone, address, id_proof
Rooms: room_id (INT) -> room_number, room_type (Single/Double/Suite), rate_per_night
Reservations: reservation_id (INT) -> check_in_date, check_out_date, total_price, booking_status,
Payments: payment_id (INT) -> payment_date, payment_method (Card/UPI/Cash), amount_paid, p
Staff: staff_id (INT) -> first_name, last_name, role (Manager/Receptionist), phone, s

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 01: Airline Reservation System - Attribute & Entity Diction

[PASTE: Exp 1: Left Page (Airline Domain Design)]

Entity Name | Primary Key | Foreign Keys | Key Attributes

Airports: airport_code (CHAR 3) -> airport_name, city, country
Aircraft: aircraft_id (INT) -> model, registration_no, total_capacity
Flights: flight_id (INT) -> flight_number, scheduled_departure, scheduled_arrival, statu
Passengers: passenger_id (INT) -> first_name, last_name, email, passport_no
Bookings: booking_id (INT) -> booking_ref (PNR), booking_date, total_amount
Tickets: ticket_id (INT) -> fare_amount, ticket_status

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 02: insert into Students (Name, ID, Course, DOB, Gender, BI

[PASTE: Exp 2: Page 11]

Number of Records: 10

Name	ID	Course	DOB	Gender	BloodGroup
Abhiram	1001	CSE-AIML	101003	M	O+ve
Abhimanyu	1002	CSE	10303	M	B+ve
Rudra	1003	CSE-DS	210503	M	O+ve
Nandani	1004	CSE	280303	F	B+ve
Ridhima	1005	CSE-BIC	31003	F	O+ve
Nithya	1006	CSE-DS	190603	F	A+ve
Rohit	1007	CSE-BIC	70203	M	A+ve
Kshithija	1008	CSE-BIC	180903	F	O+ve
Veer	1009	CSE-AIML	300503	M	B+ve
Sita	1010	CSE	291203	F	A+ve

Manual Page 11 Output (1492x739 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 02: insert into Students (Name, ID, Course, DOB, Gender, BI

[PASTE: Exp 2: Page 11]

Number of Records: 1

Name	ID	Course	DOB	Gender	BloodGroup
Ridhima	1005	CSE-BIC	31003	F	O+ve

Manual Page 11 Output (1495x204 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: alter table Student add email varchar(25);

[PASTE: Exp 3: Page 13]

Number of Records: 3

Course_Name
CSE-AIML
CSE-DS
CSE-BIC

Manual Page 13 Output (354x322 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: alter table Student add email varchar(25);

[PASTE: Exp 3: Page 13]

Number of Records: 1

sID	sName	Course_Name	Address	email
1004	Sandhya	CSE-BIC	Kurnool	null

Manual Page 13 Output (1495x208 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: select from Student WHERE sName == 'Ajay' OR Address

[PASTE: Exp 3: Page 14]

Number of Records: 2

sID	sName	Course_Name	Address	email
1002	Raj	CSE-AIML	Hyderabad	null
1004	Sandhya	CSE-BIC	Kurnool	null

Manual Page 14 Output (1480x277 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: select from Student WHERE sName == 'Ajay' OR Address

[PASTE: Exp 3: Page 14]

Number of Records: 2

sID	sName	Course_Name	Address	email
1002	Raj	CSE-AIML	Hyderabad	null
1003	Ajay	CSE-DS	Sadashivapet	null

Manual Page 14 Output (1477x274 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: select from Student WHERE sName == 'Ajay' OR Address

[PASTE: Exp 3: Page 14]

Number of Records: 3

sID	sName	Course_Name	Address	email
1002	Raj	CSE-AIML	Hyderabad	null
1004	Sandhya	CSE-BIC	Kurnool	null
1005	Ruhanika	CSE-BIC	Nandyal	null

Manual Page 14 Output (1488x309 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: select from Student WHERE sName == 'Ajay' OR Address

[PASTE: Exp 3: Page 14]

Number of Records: 3

sID	sName	Course_Name	Address	email
1002	Raj	CSE-AIML	Hyderabad	null
1003	Ajay	CSE-DS	Sadashivapet	null
1004	Sandhya	CSE-BIC	Kurnool	null

Manual Page 14 Output (1495x330 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: select from Student ORDER BY sID ASC;

[PASTE: Exp 3: Page 15]

Number of Records: 5

sID	sName	Course_Name	Address	email
1001	Anand	CSE	Secunderabad	null
1002	Raj	CSE-AIML	Hyderabad	null
1003	Ajay	CSE-DS	Sadashivapet	null
1004	Sandhya	CSE-BIC	Kurnool	null
1005	Ruhanika	CSE-BIC	Nandyal	null

Manual Page 15 Output (1477x427 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: select from Student ORDER BY sID ASC;**[PASTE: Exp 3: Page 15]**

Number of Records: 5

sID	sName	Course_Name	Address	email
1005	Ruhanika	CSE-BIC	Nandyal	null
1004	Sandhya	CSE-BIC	Kurnool	null
1003	Ajay	CSE-DS	Sadashivapet	null
1002	Raj	CSE-AIML	Hyderabad	null
1001	Anand	CSE	Secunderabad	null

Manual Page 15 Output (1488x436 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: select from Student ORDER BY sID ASC;**[PASTE: Exp 3: Page 15]**

Number of Records: 1

sID	sName	Course_Name	Address	email
1003	Ajay	CSE-DS	Sadashivapet	null

Manual Page 15 Output (1491x208 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: a. Select TOP -**[PASTE: Exp 3: Page 16]**

Number of Records: 3

sID	sName	Course_Name	Address	email
1005	Ruhanika	CSE-BIC	Nandyal	null
1004	Sandhya	CSE-BIC	Kurnool	null
1003	Ajay	CSE-DS	Sadashivapet	null

Manual Page 16 Output (1497x331 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 03: ALTER TABLE Old_table_name RENAME TO New_table_name;**[PASTE: Exp 3: Page 17]**

SQL Statement:

[Get your own SQL server](#)

SELECT TOP 3 * FROM Customers;

Edit the SQL Statement, and click "Run SQL" to see the result.

[Run SQL >](#)

Result:

Number of Records: 3

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Mexico
3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	05023	Mexico

Manual Page 17 Output (1507x891 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: Objective: ALTER Table and applying constraints:**[PASTE: Exp 4: Page 18]**

emp_no	emp_name	job	joining_date	salary	commission	dept_no
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Manual Page 18 Output (1420x57 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: insert into Employee (emp_no, emp_name, job, joining_da

[PASTE: Exp 4: Page 20]

Number of Records: 15

emp_no	emp_name	job	joining_date	salary	commission	dept_no
201001	Anand	Manager	16-08-2020	50000	1000	2
201002	Vamsi	Manager	07-05-2020	50000	1000	2
201003	Sanjana	Clerk	21-12-2020	20000	2000	4
201004	Nanditha	Clerk	30-01-2020	20000	2000	4
201005	Rahul	Salesman	19-04-2020	40000	3000	5
211001	Veer	Assistant	22-03-2021	25000	500	3
211002	Rudra	Assistant	17-05-2021	25000	500	3
211003	Kshitija	Analyst	27-11-2021	35000	1500	1
211004	Meera	Analyst	29-09-2021	35000	1500	1
211005	Rohan	Assistant	11-04-2021	25000	500	3
221001	Jaya	Manager	08-06-2022	50000	1000	2
221002	Akshay	Engineer	13-12-2022	60000	5000	6
221003	Harsha	Engineer	22-02-2022	60000	5000	6
221004	Pranav	Engineer	01-10-2022	60000	5000	6
221005	Sita	Manager	13-07-2022	50000	1000	2

Manual Page 20 Output (1488x967 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: created table

[PASTE: Exp 4: Page 21]

Number of Records: 15

salary	salary + 500
50000	50500
50000	50500
20000	20500
20000	20500
40000	40500
25000	25500
25000	25500
35000	35500
35000	35500
25000	25500
50000	50500
60000	60500
60000	60500
60000	60500
50000	50500

Manual Page 21 Output (1488x970 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: combine name and job select emp_name||job as Employee_D

[PASTE: Exp 4: Page 22]

Number of Records: 15

Employee	Annual_Salary
Anand	600000
Vamsi	600000
Sanjana	240000
Nanditha	240000
Rahul	480000
Veer	300000
Rudra	300000
Kshitija	420000
Meera	420000
Rohan	300000
Jaya	600000
Akshay	720000
Harsha	720000
Pranav	720000
Sita	600000

Manual Page 22 Output (1474x975 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: combine col with some meaning select emp_name||' works

[PASTE: Exp 4: Page 23]

Number of Records: 15

Employee_Details
AnandManager
VamsiManager
SanjanaClerk
NandithaClerk
RahulSalesman
VeerAssistant
RudraAssistant
KshitijaAnalyst
MeeraAnalyst
RohanAssistant
JayaManager
AkshayEngineer
HarshaEngineer
PranavEngineer
SitaManager

Manual Page 23 Output (1477x976 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: combine col with some meaning select emp_name||' works

[PASTE: Exp 4: Page 23]

Number of Records: 15

Employee_Details
Anand works as Manager
Vamsi works as Manager
Sanjana works as Clerk
Nanditha works as Clerk
Rahul works as Salesman
Veer works as Assistant
Rudra works as Assistant
Kshittija works as Analyst
Meera works as Analyst
Rohan works as Assistant
Jaya works as Manager
Akshay works as Engineer
Harsha works as Engineer
Pranav works as Engineer
Sita works as Manager

Manual Page 23 Output (1483x979 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: select

[PASTE: Exp 4: Page 24]

Number of Records: 1						
emp_no	emp_name	job	joining_date	salary	commission	dept_no
201005	Rahul	Salesman	19-04-2020	40000	3000	5

Manual Page 24 Output (1489x226 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: select

[PASTE: Exp 4: Page 24]

Number of Records: 4						
emp_no	emp_name	job	joining_date	salary	commission	dept_no
201001	Anand	Manager	16-08-2020	50000	1000	2
201002	Vamsi	Manager	07-05-2020	50000	1000	2
221001	Jaya	Manager	08-06-2022	50000	1000	2
221005	Sita	Manager	13-07-2022	50000	1000	2

Manual Page 24 Output (1489x373 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: display employees whose salary is between 50000 and 600

[PASTE: Exp 4: Page 25]

Number of Records: 7

emp_no	emp_name	job	joining_date	salary	commission	dept_no
201001	Anand	Manager	16-08-2020	50000	1000	2
201002	Vamsi	Manager	07-05-2020	50000	1000	2
221001	Jaya	Manager	08-06-2022	50000	1000	2
221002	Akshay	Engineer	13-12-2022	60000	5000	6
221003	Harsha	Engineer	22-02-2022	60000	5000	6
221004	Pranav	Engineer	01-10-2022	60000	5000	6
221005	Sita	Manager	13-07-2022	50000	1000	2

Manual Page 25 Output (1491x565 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: display employees whose salary is between 50000 and 600**[PASTE: Exp 4: Page 25]**

Number of Records: 11

emp_no	emp_name	job	joining_date	salary	commission	dept_no
201003	Sanjana	Clerk	21-12-2020	20000	2000	4
201004	Nanditha	Clerk	30-01-2020	20000	2000	4
201005	Rahul	Salesman	19-04-2020	40000	3000	5
211001	Veer	Assistant	22-03-2021	25000	500	3
211002	Rudra	Assistant	17-05-2021	25000	500	3
211003	Kshitiija	Analyst	27-11-2021	35000	1500	1
211004	Meera	Analyst	29-09-2021	35000	1500	1
211005	Rohan	Assistant	11-04-2021	25000	500	3
221002	Akshay	Engineer	13-12-2022	60000	5000	6
221003	Harsha	Engineer	22-02-2022	60000	5000	6
221004	Pranav	Engineer	01-10-2022	60000	5000	6

Manual Page 25 Output (1483x775 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: select emp_no, emp_name from Employee where job in ('Ma**[PASTE: Exp 4: Page 26]**

Number of Records: 6

emp_no	emp_name
201001	Anand
201002	Vamsi
201003	Sanjana
201004	Nanditha
221001	Jaya
221005	Sita

Manual Page 26 Output (1497x490 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: create table Employee(emp_no numeric(8) PRIMARY KEY, em**[PASTE: Exp 4: Page 27]**

Number of Records: 2

emp_no	emp_name
201002	Vamsi
211004	Meera

Manual Page 27 Output (1482x289 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: create table Employee(emp_no numeric(8) PRIMARY KEY, em**[PASTE: Exp 4: Page 27]**

Number of Records: 2

emp_no
201001
221002

Manual Page 27 Output (274x229 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: select emp_no from Employee where emp_name like '%r';

[PASTE: Exp 4: Page 28]

Number of Records: 1

emp_no

211001

Manual Page 28 Output (261x183 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: select emp_no from Employee where emp_name like '%r';

[PASTE: Exp 4: Page 28]

Number of Records: 15

emp_no	emp_name	job	joining_date	salary	commission	dept_no
201001	Anand	Manager	16-08-2020	50000	1000	2
201002	Vamsi	Manager	07-05-2020	50000	1000	2
201003	Sanjana	Clerk	21-12-2020	20000	2000	4
201004	Nanditha	Clerk	30-01-2020	20000	2000	4
201005	Rahul	Salesman	19-04-2020	40000	3000	5
211001	Veer	Assistant	22-03-2021	25000	500	3
211002	Rudra	Assistant	17-05-2021	25000	500	3
211003	Kshitija	Analyst	27-11-2021	35000	1500	1
211004	Meera	Analyst	29-09-2021	35000	1500	1
211005	Rohan	Assistant	11-04-2021	25000	500	3
221001	Jaya	Manager	08-06-2022	50000	1000	2
221002	Akshay	Engineer	13-12-2022	60000	5000	6
221003	Harsha	Engineer	22-02-2022	60000	5000	6
221004	Pranav	Engineer	01-10-2022	60000	5000	6
221005	Sita	Manager	13-07-2022	50000	1000	2

Manual Page 28 Output (1476x970 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: select from Employee order by emp_no desc;

[PASTE: Exp 4: Page 29]

Number of Records: 15

emp_no	emp_name	job	joining_date	salary	commission	dept_no
221005	Sita	Manager	13-07-2022	50000	1000	2
221004	Pranav	Engineer	01-10-2022	60000	5000	6
221003	Harsha	Engineer	22-02-2022	60000	5000	6
221002	Akshay	Engineer	13-12-2022	60000	5000	6
221001	Jaya	Manager	08-06-2022	50000	1000	2
211005	Rohan	Assistant	11-04-2021	25000	500	3
211004	Meera	Analyst	29-09-2021	35000	1500	1
211003	Kshitija	Analyst	27-11-2021	35000	1500	1
211002	Rudra	Assistant	17-05-2021	25000	500	3
211001	Veer	Assistant	22-03-2021	25000	500	3
201005	Rahul	Salesman	19-04-2020	40000	3000	5
201004	Nanditha	Clerk	30-01-2020	20000	2000	4
201003	Sanjana	Clerk	21-12-2020	20000	2000	4
201002	Vamsi	Manager	07-05-2020	50000	1000	2
201001	Anand	Manager	16-08-2020	50000	1000	2

Manual Page 29 Output (1483x970 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: select from Employee order by emp_no desc;

[PASTE: Exp 4: Page 29]

Number of Records: 1

min_salary

20000

Manual Page 29 Output (352x199 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: (selects the 'maximum' salary from the table)

[PASTE: Exp 4: Page 30]

Number of Records: 1

max_salary
60000

Manual Page 30 Output (352x220 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: (selects the 'maximum' salary from the table)

[PASTE: Exp 4: Page 30]

Number of Records: 1

avg_salary
40333.333333333336

Manual Page 30 Output (328x214 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: (selects the 'maximum' salary from the table)

[PASTE: Exp 4: Page 30]

Number of Records: 1

emp_name	salary
Akshay	60000

Manual Page 30 Output (1483x202 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 04: (selects the 'maximum' salary from the table)

[PASTE: Exp 4: Page 30]

Number of Records: 1

total_employees
15

Manual Page 30 Output (355x213 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: that query. An alias is created with the 'AS' keyword.

[PASTE: Exp 5: Page 31]

Number of Records: 10

Student_Name
Abhiram
Abhimanyu
Rudra
Nandani
Ridhima
Nithya
Rohit
Kshithija
Veer
Sita

Manual Page 31 Output (319x699 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: SELECT Orders.OrderID, Customers.CustomerName, Orders.O

[PASTE: Exp 5: Page 32]

Number of Records: 10

Student_ID	Blood Group
1001	O+ve
1002	B+ve
1003	O+ve
1004	B+ve
1005	B+ve
1006	A+ve
1007	A+ve
1008	O+ve
1009	B+ve
1010	A+ve

Manual Page 32 Output (1201x709 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: SELECT Orders.OrderID, Customers.CustomerName, Orders.O

[PASTE: Exp 5: Page 32]

Number of Records: 196

OrderID	CustomerName	OrderDate
10308	Ana Trujillo Emparedados y helados	9/18/1996
10365	Antonio Moreno Taquería	11/27/1996
10383	Around the Horn	12/16/1996
10355	Around the Horn	11/15/1996
10278	Berglunds snabbköp	8/12/1996
10280	Berglunds snabbköp	8/14/1996
10384	Berglunds snabbköp	12/16/1996
10436	Blondel père et fils	2/5/1997

Manual Page 32 Output (1447x526 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: select OrderID, CustomerName from Orders inner join Cus

[PASTE: Exp 5: Page 33]

Number of Records: 196

OrderID	CustomerName
10248	Wilman Kala
10249	Tradição Hipermercados
10250	Hanari Carnes
10251	Victuailles en stock
10252	Suprêmes délicies
10253	Hanari Carnes
10254	Chop-suey Chinese
10255	Richter Supermarkt
10256	Wellington Importadora
10257	HILARIÓN-Abastos

Manual Page 33 Output (822x685 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: SELECT column_name(s)

[PASTE: Exp 5: Page 34]

Number of Records: 213

CustomerName	OrderID
Alfreds Futterkiste	null
Ana Trujillo Emparedados y helados	10308
Antonio Moreno Taquería	10365
Around the Horn	10355
Around the Horn	10383
B's Beverages	10289
Berglunds snabbköp	10278
Berglunds snabbköp	10280
Berglunds snabbköp	10384
Blauer See Delikatessen	null
Blondel père et fils	10265
Blondel père et fils	10297
Blondel père et fils	10360
Blondel père et fils	10436

Manual Page 34 Output (1438x928 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: select OrderID, FirstName, LastName from Orders Right J

[PASTE: Exp 5: Page 35]

Number of Records: 196

CustomerName	OrderID
Ana Trujillo Emparedados y helados	10308
Antonio Moreno Taquería	10365
Around the Horn	10355
Around the Horn	10383
Berglunds snabbköp	10280
Berglunds snabbköp	10384
Berglunds snabbköp	10278
Blondel père et fils	10297

Manual Page 35 Output (1450x592 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: select OrderID, FirstName, LastName from Orders Right J

[PASTE: Exp 5: Page 35]

Number of Records: 197

OrderID	FirstName	LastName
null	Adam	West
10248	Steven	Buchanan
10249	Michael	Suyama
10250	Margaret	Peacock
10251	Janet	Leverling
10252	Margaret	Peacock
10253	Janet	Leverling
10254	Steven	Buchanan
10255	Anne	Dodsworth
10256	Janet	Leverling
10257	Margaret	Peacock
10258	Nancy	Davolio
10259	Margaret	Peacock
10260	Margaret	Peacock

Manual Page 35 Output (1489x916 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: select CustomerName, OrderID from Customers full join O

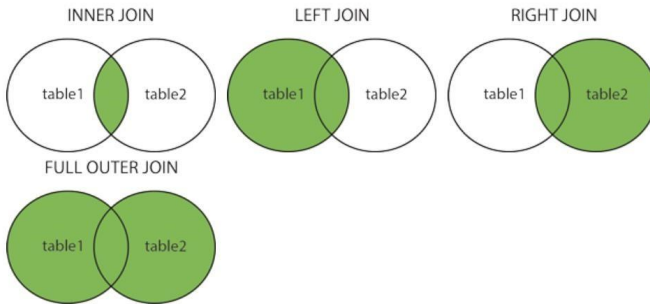
[PASTE: Exp 5: Page 36]

Number of Records: 213

CustomerName	OrderID
Alfreds Futterkiste	null
Ana Trujillo Emparedados y helados	10308
Antonio Moreno Taquería	10365
Around the Horn	10355
Around the Horn	10383
B's Beverages	10289
Berglunds snabbköp	10278
Berglunds snabbköp	10280
Berglunds snabbköp	10384
Blauer See Delikatessen	null
Blondel père et fils	10265
Blondel père et fils	10297
Blondel père et fils	10360
Blondel père et fils	10436

Manual Page 36 Output (1393x922 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: select CustomerName, OrderID from Customers full join O**[PASTE: Exp 5: Page 36]**

Manual Page 36 Output (1011x478 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: g. UNION - The UNION operator is used to combine the re**[PASTE: Exp 5: Page 37]**

Number of Records: 4

Course
CSE
CSE-AI ML
CSE-BIC
CSE-DS

Manual Page 37 Output (304x385 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 05: Manual Output #11 (Page 38)**[PASTE: Exp 5: Page 38]**

Number of Records: 14

Course
CSE-AI ML
CSE
CSE-DS
CSE
CSE
CSE-DS
CSE-BIC
CSE-BIC
CSE-AI ML
CSE
CSE-AI ML
CSE-DS
CSE-BIC
CSE-BIC

Manual Page 38 Output (337x922 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT column1, column2, ... FROM**[PASTE: Exp 6: Page 39]**

Number of Records: 10

Name	ID	Course	DOB	Gender	BloodGroup
Abhiram	1001	CSE-AI ML	101003	M	O+ve
Abhimanyu	1002	CSE	10303	M	B+ve
Rudra	1003	CSE-DS	210503	M	O+ve
Nandani	1004	CSE	280303	F	B+ve
Ridhima	1005	CSE	31003	F	B+ve
Nithya	1006	CSE-DS	190603	F	A+ve
Rohit	1007	CSE-BIC	70203	M	A+ve
Kshithija	1008	CSE-BIC	180903	F	O+ve
Veer	1009	CSE-AI ML	300503	M	B+ve
Sita	1010	CSE	291203	F	A+ve

Manual Page 39 Output (1495x733 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE '%t%';**[PASTE: Exp 6: Page 40]**

Number of Records: 4

Name	ID	Course	DOB	Gender	BloodGroup
Nithya	1006	CSE-DS	190603	F	A+ve
Rohit	1007	CSE-BIC	70203	M	A+ve
Kshithija	1008	CSE-BIC	180903	F	O+ve
Sita	1010	CSE	291203	F	A+ve

Manual Page 40 Output (1477x358 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE '%t%';**[PASTE: Exp 6: Page 40]**

Number of Records: 2

Name	ID	Course	DOB	Gender	BloodGroup
Abhiram	1001	CSE-AIML	101003	M	O+ve
Abhimanyu	1002	CSE	10303	M	B+ve

Manual Page 40 Output (1468x238 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE '%t%';**[PASTE: Exp 6: Page 40]**

Number of Records: 2

Name	ID	Course	DOB	Gender	BloodGroup
Abhiram	1001	CSE-AIML	101003	M	O+ve
Veer	1009	CSE-AIML	300503	M	B+ve

Manual Page 40 Output (1470x238 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE '%a'; Return all**[PASTE: Exp 6: Page 41]**

Number of Records: 5

Name	ID	Course	DOB	Gender	BloodGroup
Abhiram	1001	CSE-AIML	101003	M	O+ve
Abhimanyu	1002	CSE	10303	M	B+ve
Rudra	1003	CSE-DS	210503	M	O+ve
Ridhima	1005	CSE	31003	F	B+ve
Rohit	1007	CSE-BIC	70203	M	A+ve

Manual Page 41 Output (1477x412 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE '%a'; Return all**[PASTE: Exp 6: Page 41]**

Number of Records: 5

Name	ID	Course	DOB	Gender	BloodGroup
Rudra	1003	CSE-DS	210503	M	O+ve
Ridhima	1005	CSE	31003	F	B+ve
Nithya	1006	CSE-DS	190603	F	A+ve
Kshithija	1008	CSE-BIC	180903	F	O+ve
Sita	1010	CSE	291203	F	A+ve

Manual Page 41 Output (1480x412 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE '%a'; Return all**[PASTE: Exp 6: Page 41]**

Number of Records: 2

Name	ID	Course	DOB	Gender	BloodGroup
Rudra	1003	CSE-DS	210503	M	O+ve
Ridhima	1005	CSE	31003	F	B+ve

Manual Page 41 Output (1477x243 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE 'r____%';**[PASTE: Exp 6: Page 42]**

Number of Records: 2

Name	ID	Course	DOB	Gender	BloodGroup
Abhiram	1001	CSE-AIML	101003	M	O+ve
Veer	1009	CSE-AIML	300503	M	B+ve

Manual Page 42 Output (1486x249 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE 'r____%';**[PASTE: Exp 6: Page 42]**

Number of Records: 4

Name	ID	Course	DOB	Gender	BloodGroup
Abhiram	1001	CSE-AIML	101003	M	O+ve
Abhimanyu	1002	CSE	10303	M	B+ve
Rohit	1007	CSE-BIC	70203	M	A+ve
Kshithija	1008	CSE-BIC	180903	F	O+ve

Manual Page 42 Output (1480x351 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT FROM Students WHERE Name LIKE 'r____%';**[PASTE: Exp 6: Page 42]**

Number of Records: 1

Name	ID	Course	DOB	Gender	BloodGroup
Ridhima	1005	CSE	31003	F	B+ve

Manual Page 42 Output (1479x184 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: Manual Output #11 (Page 43)**[PASTE: Exp 6: Page 43]**

Number of Records: 2

Name	ID	Course	DOB	Gender	BloodGroup
Rohit	1007	CSE-BIC	70203	M	A+ve
Kshithija	1008	CSE-BIC	180903	F	O+ve

Manual Page 43 Output (1477x246 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: summary rows, like "find the number of customers in eac**[PASTE: Exp 6: Page 44]**

Number of Records: 15

emp_no	emp_name	job	joining_date	salary	commission	dept_no
201001	Anand	Manager	16-08-2020	50000	1000	2
201002	Vamsi	Manager	07-05-2020	50000	1000	2
201003	Sarjana	Clerk	21-12-2020	20000	2000	4
201004	Nanditha	Clerk	30-01-2020	20000	2000	4
201005	Rahul	Salesman	19-04-2020	40000	3000	5
211001	Veer	Assistant	22-03-2021	25000	500	3
211002	Rudra	Assistant	17-05-2021	25000	500	3
211003	Kshitija	Analyst	27-11-2021	35000	1500	1
211004	Meera	Analyst	29-09-2021	35000	1500	1
211005	Rohan	Assistant	11-04-2021	25000	500	3
221001	Jaya	Manager	08-06-2022	50000	1000	2
221002	Akshay	Engineer	13-12-2022	60000	5000	6
221003	Harsha	Engineer	22-02-2022	60000	5000	6
221004	Pranav	Engineer	01-10-2022	60000	5000	6
221005	Sita	Manager	13-07-2022	50000	1000	2

Manual Page 44 Output (1495x885 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: summary rows, like "find the number of customers in eac**[PASTE: Exp 6: Page 44]**

Number of Records: 6

COUNT(emp_no)	job
2	Analyst
3	Assistant
2	Clerk
3	Engineer
4	Manager
1	Salesman

Manual Page 44 Output (1486x478 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT COUNT(emp_no), job FROM Employee GROUP BY job OR**[PASTE: Exp 6: Page 45]**

Number of Records: 6

COUNT(emp_no)	job
4	Manager
3	Engineer
3	Assistant
2	Clerk
2	Analyst
1	Salesman

Manual Page 45 Output (1480x477 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT COUNT(emp_no), job FROM Employee GROUP BY job OR**[PASTE: Exp 6: Page 45]**

Number of Records: 5

COUNT(emp_no)	job
2	Analyst
3	Assistant
2	Clerk
3	Engineer
4	Manager

Manual Page 45 Output (1485x418 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT COUNT(emp_no), job FROM Employee GROUP BY job HA [PASTE: Exp 6: Page 46]

Number of Records: 5

COUNT(emp_no)	job
2	Analyst
2	Clerk
3	Assistant
3	Engineer
4	Manager

Manual Page 46 Output (1480x418 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: SELECT column_name(s) [PASTE: Exp 6: Page 47]

Number of Records: 24

SupplierName
Exotic Liquid
New Orleans Cajun Delights
Tokyo Traders
Mayumi's
Pavlova, Ltd.
Specialty Biscuits, Ltd.
PB Knäckebröd AB
Refrescos Americanas LTDA

Manual Page 47 Output (898x547 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: is used with SELECT, WHERE and HAVING statements [PASTE: Exp 6: Page 48]

Number of Records: 31

ProductName
Chais
Chang
Chef Anton's Cajun Seasoning
Uncle Bob's Organic Dried Pears
Konbu
Tofu
Pavlova
Teatime Chocolate Biscuits

Manual Page 48 Output (756x541 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 06: (SELECT column_name [PASTE: Exp 6: Page 49]

Number of Records: 0

ProductName

Manual Page 49 Output (583x117 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 07: CREATE VIEW - Employee Department Summary**[PASTE: Exp 7: Left Page (SQL View Output)]**

```
> CREATE VIEW View_EmpDept AS SELECT e.EmployeeID, e.Name, d.DepartmentName, e.Sal
```

EmployeeID	Name	DepartmentName	Salary
1	John Doe	HR	5000.00
2	Jane Smith	IT	6000.00
3	Emily Davis	HR	5500.00
4	Michael Brown	Finance	7000.00
5	Sarah Johnson	IT	6200.00

5 rows in set (0.01 sec)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 07: STORED PROCEDURE - GetEmployeesByDept(IN dept_id INT)**[PASTE: Exp 7: Left Page (Stored Procedure Call)]**

```
> CALL GetEmployeesByDept(101);
```

EmployeeID	Name	DepartmentID	Salary
1	John Doe	101	5000.00
3	Emily Davis	101	5500.00

2 rows in set (0.02 sec) -- Query OK

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 08: Scenario Query 1: Top Earning Employees per City**[PASTE: Exp 8: Left Page (Top Earners by City)]**

```
> SELECT City, Name, Department, Salary FROM Employees e WHERE Salary = (SELECT MA
```

City	Name	Department	Salary
New York	Alice	IT	70000.00
Chicago	Frank	Marketing	75000.00
Seattle	Grace	IT	72000.00

3 rows in set (0.02 sec)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 08: Scenario Query 2: Active Multi-Project Assignments**[PASTE: Exp 8: Left Page (Project Allocations)]**

```
> SELECT e.Name, COUNT(p.ProjectID) AS ActiveProjects, SUM(p.Budget) AS TotalBudge
```

Name	ActiveProjects	TotalBudgetManaged
Alice	2	250000.00
Frank	3	480000.00

2 rows in set (0.01 sec)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: a. Create DB (Change or Create a Database) -**[PASTE: Exp 9: Page 56]**

```
> db.createCollection("Lab8")

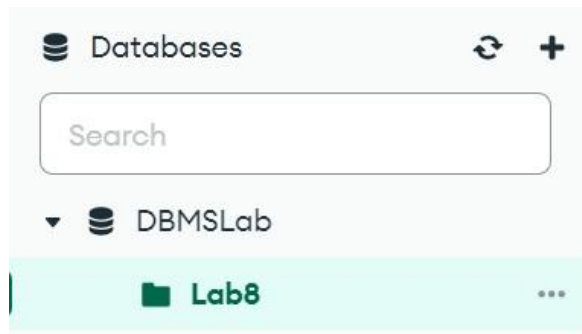
< { ok: 1 }
```

Manual Page 56 Output (361x79 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: a. Create DB (Change or Create a Database) -

[PASTE: Exp 9: Page 56]



Manual Page 56 Output (382x219 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: a. Create DB (Change or Create a Database) -

[PASTE: Exp 9: Page 56]

```
> db.Lab8.insertOne({"id":6017, "name":"Vaishnavi"})  
< {  
  acknowledged: true,  
  insertedId: ObjectId("6525814f0736eb8de184b6f7")  
}
```

Manual Page 56 Output (637x189 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: a. Create DB (Change or Create a Database) -

[PASTE: Exp 9: Page 56]

```
_id: ObjectId('6525814f0736eb8de184b6f7')  
id: 6017  
name: "Vaishnavi"
```

Manual Page 56 Output (565x133 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: a. Create DB (Change or Create a Database) -

[PASTE: Exp 9: Page 56]

```
>_MONGOSH  
  
> use DBMSLab  
  
< switched to db DBMSLab
```

Manual Page 56 Output (322x129 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: a. Create DB (Change or Create a Database) -**[PASTE: Exp 9: Page 56]**

```
> use DBMSLab  
  
< switched to db DBMSLab
```

Manual Page 56 Output (315x72 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: InsertMany() : to insert multiple values**[PASTE: Exp 9: Page 57]**

```
> db.Lab8.insertMany([{"id":6013, "name":"Vamsi"}, {"id":5037, "name":"Mohitha"}])  
  
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId("652581eb0736eb8de184b6f8"),  
    '1': ObjectId("652581eb0736eb8de184b6f9")  
  }  
}
```

Manual Page 57 Output (975x288 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: InsertMany() : to insert multiple values**[PASTE: Exp 9: Page 57]**

DBMSLab.Lab8

0 DOCUMENTS 1 INDEXES

Documents Aggregations Schema Indexes Validation

Filter Type a query: { field: 'value' } or Generate query Explain Reset Find Options

ADD DATA EXPORT DATA

1-3 of 3

<pre>{ "_id": ObjectId("6525814f9736eb8de184b6f7"), "id": 6017, "name": "Vaishnavi" }</pre>
<pre>{ "_id": ObjectId("652581eb0736eb8de184b6f8"), "id": 6013, "name": "Vamsi" }</pre>
<pre>{ "_id": ObjectId("652581eb0736eb8de184b6f9"), "id": 5037, "name": "Mohitha" }</pre>

Manual Page 57 Output (1483x778 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: InsertMany() : to insert multiple values**[PASTE: Exp 9: Page 57]**

```
> db.Lab8.insertMany([{"id":1001, "name":"John"}, {"id":1002, "name":"Smith"}])  
  
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId("65265c68e4a97f625fac3a23"),  
    '1': ObjectId("65265c68e4a97f625fac3a24")  
  }  
}
```

Manual Page 57 Output (940x286 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 09: d. Find and findOne -**[PASTE: Exp 9: Page 58]**

```
_id: ObjectId('65265c68e4a97f625fac3a23')
id: 1001
name: "John"
```

```
_id: ObjectId('65265c68e4a97f625fac3a24')
id: 1002
name: "Smith"
```

Manual Page 58 Output (873x282 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 09: d. Find and findOne -**[PASTE: Exp 9: Page 58]**

```
> db.Lab8.find()
< {
  _id: ObjectId("6525814f0736eb8de184b6f7"),
  id: 6017,
  name: 'Vaishnavi'
}
{
  _id: ObjectId("652581eb0736eb8de184b6f8"),
  id: 6013,
  name: 'Vamsi'
}
{
  _id: ObjectId("652581eb0736eb8de184b6f9"),
  id: 6037,
  name: 'Mohitha'
}
```

Manual Page 58 Output (648x588 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 09: findOne() - To select only one document**[PASTE: Exp 9: Page 59]**

```
> db.Lab8.findOne({"id":6017})
< {
  _id: ObjectId("6525814f0736eb8de184b6f7"),
  id: 6017,
  name: 'Vaishnavi'
}
```

Manual Page 59 Output (616x217 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 09: findOne() - To select only one document**[PASTE: Exp 9: Page 59]**

```
> db.Lab8.find({}, {"name":1})
< {
  _id: ObjectId("6525814f0736eb8de184b6f7"),
  name: 'Vaishnavi'
}
{
  _id: ObjectId("652581eb0736eb8de184b6f8"),
  name: 'Vamsi'
}
{
  _id: ObjectId("652581eb0736eb8de184b6f9"),
  name: 'Mohitha'
}
```

Manual Page 59 Output (595x478 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: - db.CollectionName.updateOne({reference-sub-object-sear

[PASTE: Exp 9: Page 60]

```
> db.Lab8.updateOne({"id":6017}, {$set:{"name":"Satya Vaishnavi"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 60 Output (835x292 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: - db.CollectionName.updateOne({reference-sub-object-sear

[PASTE: Exp 9: Page 60]

```
_id: ObjectId('6525814f0736eb8de184b6f7')
id: 6017
name: "Satya Vaishnavi"
```

Manual Page 60 Output (559x127 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: - db.CollectionName.updateOne({reference-sub-object-sear

[PASTE: Exp 9: Page 60]

```
> db.Lab8.updateMany({}, {$inc:{"id":1}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

Manual Page 60 Output (541x292 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: g. DeleteOne and DeleteMany - These methods accept a qu

[PASTE: Exp 9: Page 61]

```
_id: ObjectId('6525814f0736eb8de184b6f7')
id: 6019
name: "Satya Vaishnavi"
```

```
_id: ObjectId('652581eb0736eb8de184b6f8')
id: 6015
name: "Vamsi"
```

```
_id: ObjectId('652581eb0736eb8de184b6f9')
id: 5039
name: "Mohitha"
```

```
_id: ObjectId('65265c68e4a97f625fac3a23')
id: 1003
name: "John"
```

Manual Page 61 Output (583x544 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: g. DeleteOne and DeleteMany - These methods accept a qu**[PASTE: Exp 9: Page 61]**

```
> db.Lab8.deleteOne({"id":5039})  
  
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

Manual Page 61 Output (430x186 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: deleteMany() - The deleteMany() method will delete all**[PASTE: Exp 9: Page 62]**

DBMSLab.Lab8

Documents Aggregations Schema Indexes Validation

Filter Type a query: { field: 'value' } or [Generate query](#)

id: ObjectId('6520814f9730e88de3406f7f')
id: 6519
name: "Satya Valsbhavi"
id: ObjectId('6520814b0730e88de3406f7f')
id: 6519
name: "Yash"
id: ObjectId('65205c80e407f623fac3a23')
id: 1003
name: "Sai"
id: ObjectId('65205c80e407f623fac3a24')
id: 1004
name: "Sai"

Manual Page 62 Output (982x906 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: db.CollcetionName.insertOne({"subjects": ["maths", "phy**[PASTE: Exp 9: Page 63]**

```
> show dbs  
  
< admin      40.00 KiB  
  config    108.00 KiB  
  local     40.00 KiB
```

Manual Page 63 Output (259x154 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: db.CollcetionName.insertOne({"subjects": ["maths", "phy**[PASTE: Exp 9: Page 63]**

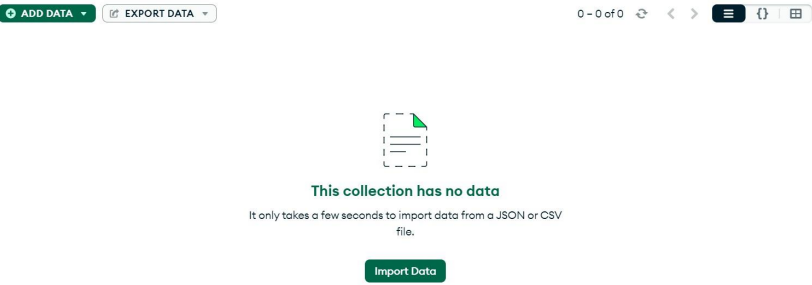
```
> db.Lab8.deleteMany({})  
  
< {  
  acknowledged: true,  
  deletedCount: 4  
}
```

Manual Page 63 Output (343x183 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 09: db.CollcetionName.insertOne({"subjects": ["maths", "phy

[PASTE: Exp 9: Page 63]

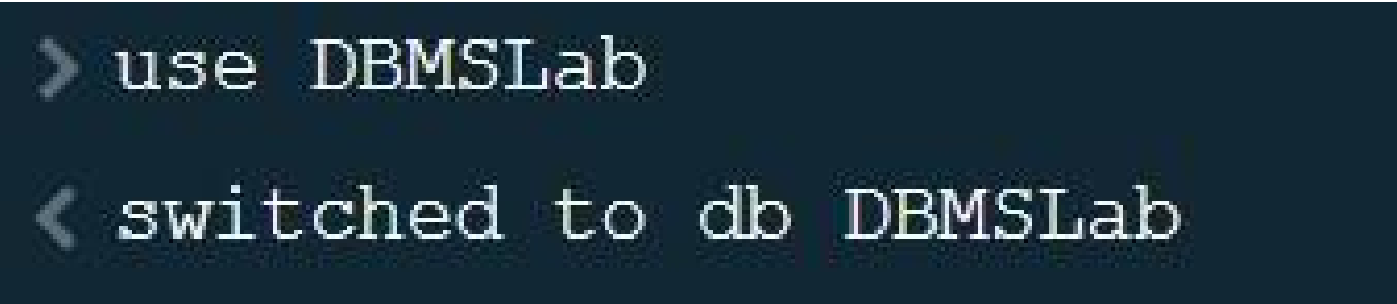


Manual Page 63 Output (1503x610 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[PASTE: Exp 10: Page 64]



Manual Page 64 Output (337x73 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[PASTE: Exp 10: Page 64]

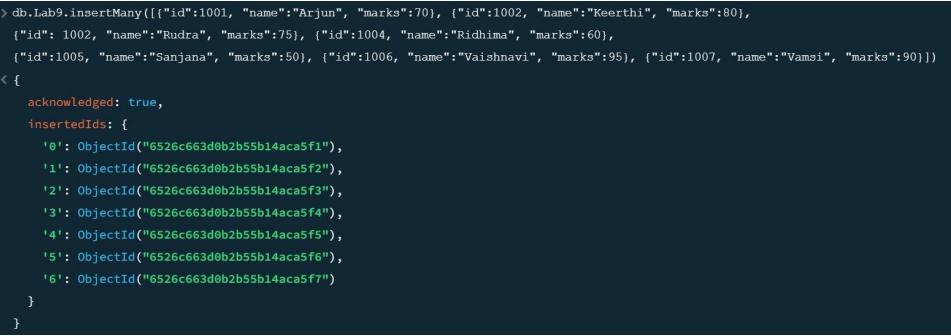


Manual Page 64 Output (364x77 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[PASTE: Exp 10: Page 64]



Manual Page 64 Output (1552x544 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[PASTE: Exp 10: Page 64]

<pre>_id: ObjectId('6526c663d0b2b55b14aca5f1') id: 1001 name: "Arjun" marks: 70</pre>
<pre>_id: ObjectId('6526c663d0b2b55b14aca5f2') id: 1002 name: "Keerthi" marks: 80</pre>
<pre>_id: ObjectId('6526c663d0b2b55b14aca5f3') id: 1003 name: "Rudra" marks: 75</pre>

Manual Page 64 Output (526x460 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Experiment No: 10

[PASTE: Exp 10: Page 64]

<pre>_id: ObjectId('6526c663d0b2b55b14aca5f4') id: 1004 name: "Ridhima" marks: 60</pre>
<pre>_id: ObjectId('6526c663d0b2b55b14aca5f5') id: 1005 name: "Sanjana" marks: 50</pre>
<pre>_id: ObjectId('6526c663d0b2b55b14aca5f6') id: 1006 name: "Vaishnavi" marks: 95</pre>

Manual Page 64 Output (558x483 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: - db.Lab9.find({"id":{\$eq: 1006}}) is equivalent to

[PASTE: Exp 10: Page 65]

<pre>_id: ObjectId('6526c663d0b2b55b14aca5f7') id: 1007 name: "Vamsi" marks: 90</pre>

Manual Page 65 Output (528x171 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: - db.Lab9.find({"id":{\$eq: 1006}}) is equivalent to

[PASTE: Exp 10: Page 65]

<pre>> db.Lab9.find({"id":{\$eq: 1006}}) < { _id: ObjectId("6526c663d0b2b55b14aca5f6"), id: 1006, name: 'Vaishnavi', marks: 95 }</pre>
--

Manual Page 65 Output (565x258 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: - db.Lab9.find({"id":{\$eq: 1006}}) is equivalent to

[PASTE: Exp 10: Page 65]

```
> db.Lab9.find({"id": 1006})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual Page 65 Output (618x256 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: selects those documents where

[PASTE: Exp 10: Page 66]

```
> db.Lab9.find({"id": {$ne: 1006}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
```

Manual Page 66 Output (595x255 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: selects those documents where

[PASTE: Exp 10: Page 66]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual Page 66 Output (577x651 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: selects those documents where

[PASTE: Exp 10: Page 66]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual Page 66 Output (562x430 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: \$gte: Value is greater than or equal to another value.**[PASTE: Exp 10: Page 67]**

```
> db.Lab9.find({"marks":{$gt: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual Page 67 Output (570x475 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: \$gte: Value is greater than or equal to another value.**[PASTE: Exp 10: Page 67]**

```
> db.Lab9.find({"marks":{$gte: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual Page 67 Output (592x685 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: \$lte: Value is less than or equal to another value. \$lt**[PASTE: Exp 10: Page 68]**

```
> db.Lab9.find({"marks":{$lt: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual Page 68 Output (559x694 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: \$lte: Value is less than or equal to another value. \$lt**[PASTE: Exp 10: Page 68]**

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual Page 68 Output (562x219 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: Manual Output #16 (Page 69)**[PASTE: Exp 10: Page 69]**

```
> db.Lab9.find({"marks":{$lte: 80}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
```

Manual Page 69 Output (571x691 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: Manual Output #17 (Page 69)**[PASTE: Exp 10: Page 69]**

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual Page 69 Output (580x433 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: \$in operator selects the**[PASTE: Exp 10: Page 70]**

```
> db.Lab9.find({"id":{"$in:[ 1001, 1005 ]}}, { _id: 0})
< {
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual Page 70 Output (642x441 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 10: \$in operator selects the**[PASTE: Exp 10: Page 70]**

```
> db.Lab9.find({"id":{"$in:[ 1001, 1005 ]}}, { _id: 1})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1")
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5")
}
```

Manual Page 70 Output (639x294 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: (<expression1>, <expression2> , and so on) and selects**[PASTE: Exp 10: Page 71]**

```
> db.Lab9.find({$and: [{ "id": {$ne: 1001}}, {"marks": {$exists: true}}]})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
```

Manual Page 71 Output (868x478 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: (<expression1>, <expression2> , and so on) and selects**[PASTE: Exp 10: Page 71]**

```
> db.Lab9.find({"id":{$in: [ 1001, 1005 ]}}, {}, { _id: 1})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual Page 71 Output (694x511 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: "id" db.Lab9.find({"id": {\$eq: 1006, \$exists:**[PASTE: Exp 10: Page 72]**

```
> db.Lab9.find({"id": {$eq: 1006, $exists: true}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual Page 72 Output (628x258 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: "id" db.Lab9.find({"id": {\$eq: 1006, \$exists:**[PASTE: Exp 10: Page 72]**

```
> db.Lab9.find({$and: [{"id": {$ne: 1001}}, {"subject": {$exists: false}}]})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual Page 72 Output (901x697 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: The query selects all documents in the

[PASTE: Exp 10: Page 73]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
```

Manual Page 73 Output (643x217 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: The query selects all documents in the

[PASTE: Exp 10: Page 73]

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual Page 73 Output (652x441 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: - db.Lab9.find({\$or: [{"marks": {\$lt: 70}}, {"id": 1006

[PASTE: Exp 10: Page 74]

```
> db.Lab9.find({"id": {$eq: 1006, $exists: true}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual Page 74 Output (628x258 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: - db.Lab9.find({\$or: [{"marks": {\$lt: 70}}, {"id": 1006

[PASTE: Exp 10: Page 74]

```
> db.Lab9.find({$or: [{"marks": {$lt: 70}}, {"id": {$eq: 1006}]})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
```

Manual Page 74 Output (790x691 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Manual Output #28 (Page 75)**[PASTE: Exp 10: Page 75]**

```
> db.Lab9.find({$or: [{"id": 1001}, {"marks": true}]})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f4"),
  id: 1004,
  name: 'Ridhima',
  marks: 60
}
```

Manual Page 75 Output (658x691 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Manual Output #29 (Page 75)**[PASTE: Exp 10: Page 75]**

```
{
  _id: ObjectId("6526c663d0b2b55b14aca5f5"),
  id: 1005,
  name: 'Sanjana',
  marks: 50
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f6"),
  id: 1006,
  name: 'Vaishnavi',
  marks: 95
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f7"),
  id: 1007,
  name: 'Vamsi',
  marks: 90
}
```

Manual Page 75 Output (612x646 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: This query will select all documents in the collection**[PASTE: Exp 10: Page 77]**

```
> db.Lab9.find({"id": {$not: {$gt: 1003 }}})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f2"),
  id: 1002,
  name: 'Keerthi',
  marks: 80
}
{
  _id: ObjectId("6526c663d0b2b55b14aca5f3"),
  id: 1003,
  name: 'Rudra',
  marks: 75
}
```

Manual Page 77 Output (579x687 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: This query will select all documents in the collection**[PASTE: Exp 10: Page 77]**

```
> db.Lab9.find({$and: [{"id": {$eq: 1001}}, {"marks": {$eq: 70}}]})
< {
  _id: ObjectId("6526c663d0b2b55b14aca5f1"),
  id: 1001,
  name: 'Arjun',
  marks: 70
}
```

Manual Page 77 Output (814x258 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: db.sampledata.find({"name":{\$regex:/^E/}})

[PASTE: Exp 10: Page 78]

```
> db.sampledata.find({"name": {$regex: /^E/}})
< {
  _id: '5f1234567890123456789015',
  employeeID: 1004,
  name: 'Emily Davis',
  age: 29,
  isEmployee: true,
  position: 'UI/UX Designer',
  salary: 70000.5,
  joinDate: '2020-06-05T00:00:00Z',
  skills: [
    'Adobe XD',
    'Sketch',
    'User Research'
  ],
  address: {
    street: '101 Pine St',
    city: 'Othercity',
    state: 'TX',
    zip: '98765'
  }
}
```

Manual Page 78 Output (637x796 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Manual Output #33 (Page 79)

[PASTE: Exp 10: Page 79]

```
{
  _id: '5f1234567890123456789022',
  employeeID: 1011,
  name: 'Emma Hernandez',
  age: 29,
  isEmployee: true,
  position: 'Product Manager',
  salary: 92000.5,
  joinDate: '2017-02-14T00:00:00Z',
  skills: [
    'Product Roadmapping',
    'Market Research',
    'User Experience'
  ],
  address: {
    street: '888 Elmwood St',
    city: 'Elsewhere',
    state: 'UT',
    zip: '12345'
  }
}
```

Manual Page 79 Output (565x759 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Manual Output #34 (Page 79)

[PASTE: Exp 10: Page 79]

```
{
  _id: '5f1234567890123456789031',
  employeeID: 1020,
  name: 'Ella Turner',
  age: 30,
  isEmployee: true,
  position: 'Sales Manager',
  salary: 88000.25,
  joinDate: '2008-01-20T00:00:00Z',
  skills: [
    'Sales Strategy',
    'Client Relationship',
    'Negotiation'
  ],
  address: {
    street: '999 Elm St',
    city: 'Nowhere',
    state: 'MA',
    zip: '99007'
  }
}
```

Manual Page 79 Output (505x763 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Manual Output #35 (Page 81)

[PASTE: Exp 10: Page 81]

```
> db.sampledata.createIndex({position: "text"})
< position_text
> db.sampledata.find(({$text:{$search: "Developer"}}))
< {
  _id: '5f1234567890123456789013',
  employeeID: 1002,
  name: 'Jane Smith',
  age: 20,
  isEmployee: true,
  position: 'Web Developer',
  salary: 65000.75,
  joinDate: '2021-03-20T00:00:00Z',
  skills: [
    'HTML',
    'CSS',
    'JavaScript'
  ],
  address: {
    street: '456 Elm St',
    city: 'Sometown',
    state: 'NY',
    zip: '67890'
  }
}
```

Manual Page 81 Output (652x838 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 10: Manual Output #36 (Page 81)**[PASTE: Exp 10: Page 81]**

```
}
{
  _id: '5f1234567890123456789015',
  employeeID: 1008,
  name: 'Ava Hall',
  age: 27,
  isEmployee: true,
  position: 'Mobile App Developer',
  salary: 72800.75,
  joinDate: '2021-01-02T08:00:00Z',
  skills: [
    'React Native',
    'Swift',
    'Android'
  ],
  address: {
    street: '555 Pineapple St',
    city: 'Somewhere',
    state: 'AZ',
    zip: '11234'
  }
}
```

Manual Page 81 Output (571x796 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: - The following updateOne() operation uses the \$inc ope**[PASTE: Exp 11: Page 83]**

```
> db.sampledata.updateOne({"name": "Emily Davis"}, {$currentDate: {"joinDate":true}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 83 Output (1029x295 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: - The following updateOne() operation uses the \$inc ope**[PASTE: Exp 11: Page 83]**

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emily Davis"
age: 29
isEmployee: true
position: "UI/UX Designer"
salary: 70000.5
joinDate: 2023-11-03T11:04:38.482+00:00
  ▶ skills: Array (3)
  ▶ address: Object
```

Manual Page 83 Output (730x316 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$rename: Renames the field. The \$rename operator update**[PASTE: Exp 11: Page 84]**

```
> db.sampledata.updateOne({"name": "Emily Davis"}, {$currentDate: {"joinDate":true}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.sampledata.updateOne({"name": "Emily Davis"}, {$inc: {"age": -1, "salary": 5000}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 84 Output (1026x579 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: \$rename: Renames the field. The \$rename operator update**[PASTE: Exp 11: Page 84]**

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emily Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 75000.5
joinDate: 2023-11-03T11:04:38.482+00:00
▸ skills: Array (3)
▸ address: Object
```

Manual Page 84 Output (604x292 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - db.sampledata.updateOne({"name":"Emily Davis"}, {\$set**[PASTE: Exp 11: Page 85]**

```
> db.sampledata.updateOne({"name":"Emily Davis"}, {$set: {"name":"Emilia Davis", "salary":10000}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 85 Output (1198x292 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - db.sampledata.updateOne({"name":"Emily Davis"}, {\$set**[PASTE: Exp 11: Page 85]**

```
> db.sampledata.updateOne({"name":"Emily Davis"}, {$rename: {"joinDate":"JoiningDate"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 85 Output (1039x289 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - db.sampledata.updateOne({"name":"Emily Davis"}, {\$set**[PASTE: Exp 11: Page 85]**

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emily Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 75000.5
▸ skills: Array (3)
▸ address: Object
JoiningDate: 2023-11-03T11:04:38.482+00:00
```

Manual Page 85 Output (577x319 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: - db.sampledata.updateOne({"name": "Sophia Brown"}, {\$s**[PASTE: Exp 11: Page 86]**

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emilia Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 10000
▶ skills: Array (3)
▶ address: Object
JoiningDate: 2023-11-03T11:04:38.482+00:00
```

Manual Page 86 Output (627x312 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: 'new value only' like shown in the value.**[PASTE: Exp 11: Page 87]**

```
> db.sampledata.updateOne({"name":"Sophia Brown"}, {$set: {"skills":["Python","R","Machine Learning","C++"]}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 87 Output (1333x301 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: 'new value only' like shown in the value.**[PASTE: Exp 11: Page 87]**

```
_id: "5f1234567890123456789017"
employeeID: 1006
name: "Sophia Brown"
age: 31
isEmployee: true
position: "Data Scientist"
salary: 90000
joinDate: "2017-07-20T00:00:00Z"
▼ skills: Array (4)
  0: "Python"
  1: "R"
  2: "Machine Learning"
  3: "C++"
▼ address: Object
  street: "333 Cedar St"
  city: "Othertown"
  state: "IL"
  zip: "33445"
```

Manual Page 87 Output (478x520 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: \$unset: Removes the field from the document. The \$unset**[PASTE: Exp 11: Page 88]**

```
> db.sampledata.updateOne({"name":"Sophia Brown"}, {$set: {"skills":["C++"]}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 88 Output (994x298 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: \$unset: Removes the field from the document. The \$unset**[PASTE: Exp 11: Page 88]**

```
_id: "5f1234567890123456789017"
employeeID: 1006
name: "Sophia Brown"
age: 31
isEmployee: true
position: "Data Scientist"
salary: 90000
joinDate: "2017-07-20T00:00:00Z"
▼ skills: Array (1)
  0: "C++"
▼ address: Object
  street: "333 Cedar St"
  city: "Othertown"
  state: "IL"
  zip: "33445"
```

Manual Page 88 Output (529x433 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - db.sampledata.updateOne({"name": "Ava Hall"}, {\$addTo**[PASTE: Exp 11: Page 89]**

```
> db.sampledata.updateOne({"name": "Emilia Davis"}, {$unset: {"JoiningDate": " ", "address": " "}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 89 Output (1147x297 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - db.sampledata.updateOne({"name": "Ava Hall"}, {\$addTo**[PASTE: Exp 11: Page 89]**

```
_id: "5f1234567890123456789015"
employeeID: 1004
name: "Emilia Davis"
age: 28
isEmployee: true
position: "UI/UX Designer"
salary: 10000
▼ skills: Array (3)
  0: "Adobe XD"
  1: "Sketch"
  2: "User Research"
```

Manual Page 89 Output (490x330 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - db.sampledata.updateOne({"name": "Ava Hall"}, {\$pop:**[PASTE: Exp 11: Page 90]**

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$addToSet: {"skills": "Java"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 90 Output (994x295 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - db.sampledata.updateOne({"name": "Ava Hall"}, {\$pop:**[PASTE: Exp 11: Page 90]**

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
▼ skills: Array (4)
  0: "React Native"
  1: "Swift"
  2: "Android"
  3: "Java"
▼ address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual Page 90 Output (480x489 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: of the fields can differ.**[PASTE: Exp 11: Page 91]**

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$pop: {"skills": -1}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 91 Output (928x300 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: of the fields can differ.**[PASTE: Exp 11: Page 91]**

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
▼ skills: Array (3)
  0: "Swift"
  1: "Android"
  2: "Java"
▼ address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual Page 91 Output (523x484 px)

-- [CUT] CUT ALONG DASHED LINE [CUT] --

EXP 11: - {\$pull: {<field1>: <value|condition>, <field2>: <valu**[PASTE: Exp 11: Page 92]**

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$pull: {"skills": {$in: ["Swift", "Android", "Java"]}}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 92 Output (1261x300 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: - {\$pull: {<field1>: <value|condition>, <field2>: <valu

[PASTE: Exp 11: Page 92]

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
skills: Array (empty)
address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual Page 92 Output (454x400 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: Manual Output #21 (Page 93)

[PASTE: Exp 11: Page 93]

```
> db.sampledata.updateOne({"name": "Ava Hall"}, {$push: {"skills": ["React Native", "Swift", "Android", "JavaScript", "C++"]}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Manual Page 93 Output (1518x285 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 11: Manual Output #22 (Page 93)

[PASTE: Exp 11: Page 93]

```
_id: "5f1234567890123456789019"
employeeID: 1008
name: "Ava Hall"
age: 27
isEmployee: true
position: "Mobile App Developer"
salary: 72000.75
joinDate: "2021-01-02T00:00:00Z"
skills: Array (1)
  0: Array (5)
    0: "React Native"
    1: "Swift"
    2: "Android"
    3: "JavaScript"
    4: "C++"
address: Object
  street: "555 Pineapple St"
  city: "Somewhere"
  state: "AZ"
  zip: "11234"
```

Manual Page 93 Output (459x532 px)

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 12: \$match and \$group - Total Sales by Product Category

[PASTE: Exp 12: Left Page (Aggregation \$group)]

```
> db.sales.aggregate([ { $match: { status: 'Completed' } }, { $group: { _id: '$cat
[
  { "_id": "Electronics", "totalRevenue": 450000, "count": 28 },
  { "_id": "Furniture", "totalRevenue": 185000, "count": 14 },
  { "_id": "Stationery", "totalRevenue": 32000, "count": 45 }
]
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 12: \$lookup Stage - Left Outer Join Across Collections

[PASTE: Exp 12: Left Page (Aggregation \$lookup)]

```
> db.orders.aggregate([ { $lookup: { from: 'customers', localField: 'customer_id',
[
  { "_id": ObjectId("60a1b..."), "order_no": "ORD-1001", "total": 12500, "customer_name": "Aarav Sharma" },
  { "_id": ObjectId("60a1c..."), "order_no": "ORD-1002", "total": 4800, "customer_name": "Priya Patel" }
]
```

[CUT] CUT ALONG DASHED LINE [CUT]

EXP 13: Creating Compound & Full-Text Indexes

[PASTE: Exp 13: Left Page (Index Creation)]

```
> db.articles.createIndex({ content: 'text', title: 'text' }); db.employees.create
{
  "numIndexesBefore": 1,
  "numIndexesAfter": 2,
  "createdCollectionAutomatically": false,
  "ok": 1
}
```


[CUT] CUT ALONG DASHED LINE [CUT]

EXP 13: Full-Text Search Query Execution (\$text / \$search)

[PASTE: Exp 13: Left Page (Text Search Results)]

```
> db.articles.find({ $text: { $search: 'database NoSQL' } }, { score: { $meta: 'te
[
  { "_id": ObjectId("61a..."), "title": "Intro to NoSQL", "content": "MongoDB is a scalable NoSQL database...", "score": 1.5 },
  { "_id": ObjectId("61b..."), "title": "Modern DBs", "content": "Databases like MongoDB power cloud apps...", "score": 1.1 }
]
```